

EX380v4.14-Certified Specialist in OpenShift Automation and Integration

Q1: Configure LDAP Authentication

in Exam

<u>Key</u>	<u>Value</u>
Name	LDAP-AUTH-EX380
Connection Protocol	ldaps
Server Hostname	ldaps:// auth.domain21.example.com/dc=domain21,dc=example,dc=com?uid
Base DN	cn=users, cn=accounts, dc=domain21, dc=example, dc=com
Bind DN	uid=admin, cn=users, cn=accounts, dc=domain21, dc=example, dc=com
Bind Password	Oompotia21
Insecure	false
Query filter	uid
Mapping Method:	claim

Given the OAuth information above configure the “LDAP authentication type” according to the following requirements:

- The project **openshift-config** contains a generic secret named **ldap-auth-secret** with the bind password
- The project **openshift-config** contains a configuration map named **ldap-auth-ca** with content from <https://auth.domain21.example.com/ipa/config/ca.crt> and preserves the original name of the certificate.
- The attributes id=dn, email=mail, name=cn and preferredUsername=uid are configured
- username **rmorales** must be able to login using the password **Oompotia21**

in Lab to practice

Guided Exercise ->DAP Authentication and Group Synchronization **page 27**

LDAP search query

<u>Key</u>	<u>Value</u>
Base DN (-D)	cn=Directory Manager
URI (-H)	ldaps://rhds.ocp4.example.com
Password (-w)	redhatocp

LADAP validated information

Name	Red Hat Directory Server
Connection Protocol	ldaps
URL & BaseDN	ldaps:// rhds.ocp4.example.com/dc=example,dc=com?uid
Bind DN	cn=Directory Manager
Bind password	redhatocp
Email	mail
Insecure	false
Query filter	uid
Mapping Method:	claim
Server Hostname	auth.domain21.example.com

```
$ lab start auth-ldap
```

Q2: Create and use a Client Certificate

The ACME Security company has been hired to audit your OpenShift cluster.

In preparation for this audit, you need to create a client certificate that allows the auditor to examine everything in the cluster but does not allow the auditor to make any changes.

Configure your OpenShift cluster for this audit with the following requirements:

- A client certificate exists with the username: **acme-auditor**
- A group exists with the name: **acme-corp**
- Members of this group have access to the cluster role: **cluster-reader**

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- The client certificate exists in **/home/opsadm/kubeconfig-acme.config**
- The client certificate must not be able to **create** or **delete** projects
- The client certificate must be able to **view** all pods in the cluster

Solution

Create a client certificate for a backdoor administrator account to use if the IdP that provides the administrator account fails.

The username for the administrator certificate must be **acme-auditor**, and the user must be part of the backdoor **acme-corp** group with the **cluster-reader** cluster role. Set the expiration time for the certificate to one year.

```
username: acme-auditor
```

group: acme-corp

cluster-role: cluster-reader[illegible]

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```
RVMgp3S1hQcXpsVUIDS2tiem02aXlrSXpMck5rMzBVQ1cvNGJrb1I5SIQ5TDZpblAwVE1qeWxieVYzaFlsU1FYdmJuCnJzVG1RVX
ZXSmRuWjICVm12YkVCT1c5a1o1bmdUTG5GeHVzcmtncHRxYk5RSnZCMFBWN3UwQytkMElWdHpmSzYKbkdGU2lkU2tLcWN
DWjQvSVUxRzZKQnZvVzJmNnpMNEU4eDZVS2FnQ2IzRjU3L0pnSEJjME1zczRRREhmbGEzLwpaRE02bENBOUJTL1M0QzNiZW
VSMFVJWmFRbkdVcURKdXhWVlhhcDFyb0RHMzMmUFBvVm5zTnFsVTVKTVp6U2U0CkloZ1JESkp4YkgrRWpZQXRVMXI3cDAX
MmtCeFI1RWRSV3Q0a3IJRkt6RFA4RkcxTE1RVXNpZUZ3dCt4azNQK2oKMnhCbWRuZHQ3SWc3QzBYdThQY1VreC9JZFhKK3Az
c1JrM01jNWD6cjZ6dzVGR0xockQ2KzZ3Szg0NWFNaWRubgpGd0ZRBldrMi9zOEFqMmtOdkhodUxRQ1dsV0Q0eno4TENqaEZQ
WTJMVfG2cjhyNXZqZG5OT0p3aGRRS29neThRCnR4RjVZeFg3dnVxaDhaZGNod0crWGFEb1NON0ZRYjRBazdiYks0eHZGNEdC
WkdvSgotLS0tLUVORCBDRVJUSUZJQ0FURSBSRVFVRVNULS0tLS0K
student@workstation acme]$ cat acme-cert-sign-request.yaml
kind: CertificateSigningRequest
apiVersion: certificates.k8s.io/v1
metadata:
  name: acme-cert-sign-req
spec:
  signerName: kubernetes.io/kube-apiserver-client
  expirationSeconds: 86400
  request: LS0tLS1CRUdJTiBDRVJUSUZJQ0FURSBSRVFVRVNULS0tLS0KTUJlRWNEQ0NBbGdDQVFBd0t6RVNNQkFHQTFRUNn
d0pZV050WIMxamNtOXNdNUIV3RXdZRFZRUUREQXhoWTIxbApMV0YxWkdsMGZlSXdnZ0lpTUeWR0NTcUdTSWlZRFFFQkFRVU
FBNEIDRHdBd2dnSUtBb0lDQVFDenA4aER6WGtQCnhDZTlwUXIMYlJ0aTB0em1WSTN3VmRPUHlZ2Zr2VXJndXNwTy9lak9leX
dRMHB4NVeYa1JkWKrLeU85SEJ1S3YKEdBpQXIjTajhYakd5Qk6MjZsWGhKWEZVQzVjY1ZjK1RKcjVZYjBsQ0R4M1c4SXJqQ290
T0VOeFpDOURSeVIEagpKcVJ1RUN1NctIWUVmdkUrMjJkGvYqYUixZW9DK1IMM1BhY1k5SGN0aXlZn29wYm5TVk9NZXJ3WH
gwaldVaTMrCmp2OENneUVMG9EU2l1MTVUWWNaNEpMMVNUL3BwOFUrcytwclDhZWxhRUVqQjR6Y1pMd0N3T09Fb2RO
QmdEMVcKcVpHdUFUNElwVmRjT25NeTVBTnRnUWp6R2ZmSWsrTm9Jcm1qZEdxcDExN0Qwak5kQXJHd2F0d2QyVUdFTGFN
dgowNkc3bDViWVFTNlRTT0F1UFppU0gyQnlnK2tkalVZNVUyYndvUTlzMVhXm2JpNlJXNk9NNElsRDI3ZVhYYnpGCKi3TKx6OVI1
aVoxR0xNSHovY3NLeHFUMVBBZDQ2QkRReUowTHUrNHNmS2lT3lDaGrSteh4eVR0NWFcSkJFeTQKZWVONE9qZlSQkF3Rn
YzLzUySEWycE9tMFJjbGFaSnhQMlBvRXo4MnRJamcybDZwT0JzRVUxdEthdXVBZHIjQcApNZDFvVlg1Z3YrU3o1TGdvM3JLK1d0
VEUzYW1ReHhTK2IzZHBETZjMXy4SzREK2pTOUIVa0dvMXyrK2RMNHRkCng0cG1IQ3hFUVZqUWkydUd2aDA3VzM5NmZBaT
RsMWpFL29xZnFMQzBtaERVUG51c0xQcWhjWmdjd3dMTkRLNIEKvjXV0J0cVM4ZkpSN2ZYQkVkrK4yNWZpdUF2cUowblNP
d0IEQVFBQm9BQXdeUUVKS29aSWWh2Y05BUUVMQlFBRApnZ0ICQUhVcTQxbktHT2l4THcyUENXZmlFZDUzTTE2cmt5UjFhWFNZ
L2xnT1c4MTA5OEKvSVhGS0FTQUFsUWtYcjl0Q2tUOUZJUklpcFBPQVNEZ0tuUk5pVGZoNGgvd1gvdnVESmVOQkNOSHpxbUo
zdIjZQlFYcEtZRzJzRDFGZ2kKbjVMMW9HR0RzMis0SzhpNkhoVDR4R2lYmms5QkZiNHBRTGdtUWZPY2dITUY5L3VSRWpqdkZC
ejBFRStXSIRVMgp3S1hQcXpsVUIDS2tiem02aXlrSXpMck5rMzBVQ1cvNGJrb1I5SIQ5TDZpblAwVE1qeWxieVYzaFlsU1FYdmJuC
nJzVG1RVXZXSmRuWjICVm12YkVCT1c5a1o1bmdUTG5GeHVzcmtncHRxYk5RSnZCMFBWN3UwQytkMElWdHpmSzYKbkdGU2
lkU2tLcWNWDWjQvSVUxRzZKQnZvVzJmNnpMNEU4eDZVS2FnQ2IzRjU3L0pnSEJjME1zczRRREhmbGEzLwpaRE02bENBOUJTL1
M0QzNiZWVSMFVJWmFRbkdVcURKdXhWVlhhcDFyb0RHMzMmUFBvVm5zTnFsVTVKTVp6U2U0CkloZ1JESkp4YkgrRWpZQXR
VMXI3cDAXMmtCeFI1RWRSV3Q0a3IJRkt6RFA4RkcxTE1RVXNpZUZ3dCt4azNQK2oKMnhCbWRuZHQ3SWc3QzBYdThQY1VreC
9JZFhKK3AzC1JrM01jNWD6cjZ6dzVGR0xockQ2KzZ3Szg0NWFNaWRubgpGd0ZRBldrMi9zOEFqMmtOdkhodUxRQ1dsV0Q0eno
4TENqaEZQWTJMVfG2cjhyNXZqZG5OT0p3aGRRS29neThRCnR4RjVZeFg3dnVxaDhaZGNod0crWGFEb1NON0ZRYjRBazdiYks0
eHZGNEdCWkdvSgotLS0tLUVORCBDRVJUSUZJQ0FURSBSRVFVRVNULS0tLS0K
usages:
  - client auth
:wq!
[student@workstation acme]$ oc create -f acme-cert-sign-request.yaml
certificatesigningrequest.certificates.k8s.io/acme-cert-sign-req created

[student@workstation acme]$ oc get csr
NAME          AGE  SIGNERNAME                                REQUESTOR    REQUESTEDDURATION   CONDITION
acme-cert-sign-req  44s  kubernetes.io/kube-apiserver-client  admin        24h                Pending
[student@workstation acme]$ oc adm certificate approve acme-cert-sign-req
certificatesigningrequest.certificates.k8s.io/acme-cert-sign-req approved
```

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```
[student@workstation acme]$ oc get csr
NAME          AGE  SIGNERNAME                                REQUESTOR   REQUESTEDDURATION   CONDITION
acme-cert-sign-req  2m5s  kubernetes.io/kube-apiserver-client  admin      24h                  Approved,Issued
[student@workstation acme]$ oc get csr acme-cert-sign-req -o jsonpath='{.status.certificate}' | base64 -d >acme-cert-sign-req-access.crt
[student@workstation acme]$ ls
acme-auditor-auth.csr  acme-cert-sign-req-access.crt  acme-cert-sign-request.yaml  tls.key
[student@workstation acme]$ oc config --kubeconfig acme-corp.config set-credentials acme-auditor --client-certificate
acme-cert-sign-req-access.crt --client-key tls.key --embed-certs
User "acme-auditor" set.
[student@workstation acme]$ cat acme-corp.config
apiVersion: v1
clusters: null
contexts: null
current-context: ""
kind: Config
preferences: {}
users:
- name: acme-auditor
  user:
    client-certificate-data:
LS0tLS1CRUdJTiBDRVJUSUZJQ0FUR50tLS0tCk1JSUVJakNDQXdzZ0F3SUJBZ0lSQU1NSkhERTZzUVR5cTVMZW53VHdOWTh3RF
FZSkvWklodmNOQVFFTEJRQXcKSmPfa01DSUdBMVVFQXd3YmEzVmllaUzFqYzNjdGM5bG5ibVZ5WDBBeE56TTJNak16TmPj
MU1CNfHEVEkxTURFdwpOekE0TIRVeU4xb1hEVEkxTURFd09EQTROVFV5TjFvd0t6RVNNQkFHQTFRUNoTUpZV050WIMxam
NtOXdnUIV3CkV3WURWUUVFERXd4aFkyMWxMV0YxWkdsMGlzSXdnZ0lpTUeWRONTcUdTSWlZrFFFQkFRVUFBNEDRhdBd2d
nSUSkQW9JQ0FRQ3pwOGhEelhrUUhDZTlwUXIMYlJ0aTB0em1WSTN3VmRPUHl2ZzR2VXJndXNwTy9lak9leXdRMHBB4NQpRM
mtSZFpES3IPOUhCdUt2eDBpQXlTajhYakd5Qk16MjZsWGhKWEZVQzVjY1ZjK1RKcjVZYjBsQ0R4M1c4SXJqCkNvTk9FTnhaQzIEU
nIzRGpKcVJ1RUN1NctIWUWmdkUrmJjkCgVqYUixZW9DK1IMM1BhY1k5SGN0aXlzN29wYm4KU1ZPTWVyd1h4MGpXVWkzK2
p2OENneU1VMG9EU2l1MTVUWwNANePMMVNul3BwOFUrcytwcldhZWxhRUVqQjR6YwpaTHdDd09PRW9kTkJnRDFXcVpHd
UFUNElwVmRjT25NeTVBTnRnUWp6R2ZmSWsrTm9Jcm1qZEexcDExN0Qwak5kCkFyR3dhdHdkMIVHRUxhTXYwNkc3bDViWV
FTNIRTT0F1UFppU0gyQnlnK2tkalVZNVUyYndvUTlzMVhXm2JpNlIKVzZPTTRJbEQyN2VYWGJ6RkI3Tkx6OVI1aVoxR0xNSHovY3
NLeHFUMVBBZDQ2QkRReUowTHUrNHNmS2lIT3lDaApkbExleHIUdDVhQkpCRXk0ZWVONE9qQzISQkF3RnYzLzUySEwycE9tM
FjBGFaSnhQMlBvRXo4MnRJamcybDZwCk9Cc0VVMXRLYXV1QWRyUHBNDZDFvVlg1Z3YrU3o1TGdV3JLK1d0VEUzYW1ReHhT
K2lzZHBEUTZjMXY4S4zREK2oKUZlJVWtHbzF2KytkTDR0ZHg0cG1IQ3hFUVZqUWkydUd2aDA3VzM5NmZBaTRsMWpFL29xZnFM
QzBtaERVUG51cwpMQkFoY1pnY3d3TE5ESzZRVjJXV0J0cVM4ZkpSN2ZYQkVKKRk4yNWZpdUF2cUowbINP0IEQVFBQm8wWX
dSREFUCKJnTlZlU1VFRERBS0JnZ3JCZ0VGQlFjREFqQU1CZ05WSFJNQkFmOEVBakFBTUI4R0ExVWRJd1FZTUJhQUZOMEsKT2ZT
RTFBZmFUY2krTjF2UzVJRmM0ZFpKTUeWRONTcUdTSWlZrFFFQkN3VUFBNEDRhdBd2d09PRW9kTkJnRDFXcVpHdQpBVDRJcFZkY09u
TXk1QU50Z1FqekdmZklrK05vSXJtamRHcXAMtDEMGP0ZEFyR3dhdHdkMIVHRUxhTXYwNkc3Cmw1YlIRUzZUU09BdVBaaVNI
MkJ5ZytrZGpVWTVVMmJ3b1E5czFYcTNiaTZSVzZPTTRJbEQyN2VYWGJ6RkI3TkWKejlZdWlaMUdMTUh6L2NzS3hxVDFQQWQ0
NkJEUxIKMEx1KzRzZktpZU95Q2hkbExleHIUdDVhQkpCRXk0ZWVONApPakM5Uk1Bd0Z2My81MkhMMnBPbTBSY2xhWkp4UD
```

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JQb0V6ODJ0SWpnMmw2cE9Cc0VVMXRLYXV1QWRyUHBZNDFvCIZYNWd2K1N6NUxbzNySytXdFRFM2FtUXh4Uytpc2RwRFE
2YzF2OEs0RCtqUzIJVWtHbzF2KytkTDR0ZHg0cG0KSEN4RVFWaIFpMnVHdmgwN1czOTZmQWk0bDFqRS9vcWZxTEMwbWhEV
VBudXNMQkFoY1pnY3d3TE5ESzZRVjJXVwpCdHFTOGZKUjdmWEJFSkZOMjVmaXVBdnFKMG5TT3dJREFRQUJBb0IDQUFiSmRn
b3RzZfJKSmthR3A0aUFPTjBpCllkV252Y0dsdjBFcVhKSUs2MUJvYmhVa01zNStIbVkbz1ZGVUHSemJrcVUraUYrU3RkbjF6dG8zL3
o3aXkydDgKdHFBVfUdzJDYjBtNmtZQ21JMW9GTXAxZXIYWTVBUGNqQ1ZUU1M1Q3BTTU9ER0pUTGtzbWIMd0p0Z1VqdEh
NQQpzcTl4RFhhUmR6YnVkl3MxeVluNmxSeWZFUDR5VFPInJLQnAzT2luM2F3NDIlK2hlaVd0bEVVQzNEVHFLNU5kCjlwRUIh
ZXdETFE5c05EOHpweTF5WnVRNFVRVGNLk2h5dEJKdFp5R08xT21MK0NyM2VSbSszVEhlejgraGVNRSsKQ2dwb1RSZ3V4dWp
4Z3dnR1RPY3JES3FpcEJiczBTS0NYSTROSXl1cTFjeExpXmVzlhTUffRTVJFU2RKU3hwQQpETm5ZOGVhZWtpRS93dFhmclVvbW
xLeEU1SWttcEN1eXI0ckxmSHZ3ampNbHhoT0ZQcXpiRjF2aFiTHFEQVQvCnVDY0drRDRFYjBNcU9reXBpQkFRb25ydUNkeks2a
U5TSIIeQWtQM1NKZ21EN0JwVy9wdjV0NIBIVHdCQ2R3TW8Kb21ubjZNUtRHZCtna2Z6MVF5VnIIRTByZ2VYbTdobIRqUVhMN
TdnNlovc3VxcDhkUEdBTUpwZjY4eVN4Y0F5aAp3QksrRnB1YU13a2lpNHplYXIYM2xEODIwejdRsk13SVhHRG1PQW5sVExucktx
cG9NRjJ2RENHdmpUTDhkVzFpCnlWUUFM4NHpZSEpkSkxpOUYzd0pzVncyTEY2Y1I5VWZocGlqU1oxMEEXLRkpMRzVpSU5wRU0
xd1sdEV6dmsvYlckS1RKQ2QzS25EZWpuZzdZOE NmSnBBb0lCQVFERNLGR5JRXdON09BMDZyQW1kMDNNT2RYdDVjN3FQb
1FzaFZBTApbjY3OFcyVndBdmptQitIVnR0OUxnRWWaK256eW9mR3M5THN6dkdpU0JET0V3eDFNQINvL09maU4yK3luaTYyC
mpEeWIEZ05odEFwZjQ2bXF5UXFuVeliUU9IMVE2YWR3NFZ5Nm01bFpCRGNDMWIHYzgybFRVOWFhbWxMYXpMRmQKbmZ
QOXDZQXlsNW1qWWtoSGtiTy8yS3VVbVZBYWV3K3k3ZkV4ekRQL1BiWEhwMnhoa0ZBNVUzRU1qNWFhaWFqTApzSi9GTmt1
NmJXUzVkrFEvQ1BXUzIzeU9OVEVjb1dwVWdEM3F2RGRIVVBhdUtNcjJ6dGIIYjZnSzdrRjTbXpUCjRGNU9VaHpsVitKU0huWEJ
qb2RVSzRPK3FTVjNRdkVoNUK5UIE0VFNhY2ILRGcyakFvSUJBUURxMFVhckJtUEQKbFVPUVdkR1BXOFNjaStLdlcySEtma0V5SUZ
zdFE2ZnRyRzFPN0lrTDZGcXFENEw5NVJSTIB5VGNvZDVsekxGMwpCcXVrb0NQQk14T2M1MUNlcG5uVFRVS3pnZXlxejl4MFFh
d01pUzJXcFhJKzNjQUtUbXfmbVrtZi9JclJ6WW9rCm9TeGtMTUxQUzIOQUlZr25xa3QwMWtPRWlsVVh4TEJNTkViaktmN3VxM
HhpSjlqQXJ1VTkxR01SMXNBuY9yNUSKYXRBK5pR056bkV4R21oaXM5Qzk0TkJEctFYl3Y5WfJBOWhpdjhVbXdUdXdwV0p3R
1JRTFF3RGRYdGI4UTVaNAo1MDY0Y09ZdVhmRWFdYVpXVkJuSmdKU0RPSDBJY2tWVv5rNFIUbjlyZUNPKzd3ZTZR1XIVbk1PST
RFeUwzMC9ECngxNWFWeEpyeGNLSkFvSUJBUUREYVJvM3ZxZGI0VIMwTnZwc1JsZWVUSTI5L3F4RVZQODJGQVBhcjllRjE4aEg
Kd2dXYW15bkpzOFRemNOYi9MMFQRsnNpaWtNRTITdHJoNDhNcTBDZ29zbmFHQW1hVDBqbjRwNGtqeUhuWINmRwprUVJ
ReXlsNk5YNmxqOVZzVkg4TFhUTXJrZzBJT1NYMEJCcUdmL0dsVDGxano1K3JQqkV5TDBzK1pkc0s3OFhoCnRCTWJTWEZueGx5Y
UthaDJaOGZkdZyZVURVVzVNMDbSvZFJbUlrb2x4UC9sVWZUV1hXUWE1d0U3eUFwL2pqNWIKejh6UW9ybUZzb0NnZHM5OG
FtR0hUnhIRm5Pd0FQMfU0cDViQUdRUGZpdldaSFk2QVovYVQwTG8yVS85d0RzSQpMbmlsQ2JMZTNENzBvVDJIT2RBaGZ1NX
ByRW01QUM1Q2FEd21VUU1WQW9JQkFRRGw4NmXqEc4Z1VtVUcwSjZpCjZEaERJYzRNOGMxeIVtNmdHd1dPN3lsZFhEVHp
TOUtjM1I4a1dPcHJudeZoMEV0VjhrSnM2S2FGckRIY2tPSIMKeWfVVM0WVdRVIIITWhQMjNHNGEzRkhFVzZIVDR5YU0xMXB
vN0JtclBoMjJTNFFIQTE4aWxFa0xFL2IzNXNtagozVnB6NHBAUG5qb2tvY3NIMmFUNUVMbWc5clRRc3dpZIM0c1N1SGpRUUxY
bjlieEs5T1hraXBoZElmcXlqODhPCKlyUEFaVjhdQVp1U3hyek05OVhSRFZqalJnTzZ1SmZtR3VkTUNHSE02eEI3OFIzQXE2YVhCYz
B1ZXo0ZzZ4YIAKSTFuL3FLR0creld1V2NIV05WUmtXQnREbGxtbC9nRIVDaVBmVUE5ZXNDcjJnZ29PV2JFOGJLU2RVZms1T1Btb
QpKdFVaQW9JQkFHZTRkeTc2eGhpY0dqSWNkRDZwYkRic2JwK3pkRnJNUXFQK3dIZkwzOGdIT0E3eFdPdInlcXIVCkE4YUwwRis
1WkNGTmNLdWlqbUJBV0hzenVYNzBBEWJzcDlhZWWhnaThMWGIQYIJMR0VktX5d5dZhenkyZnFhZDYKWfhdTNCdnkyTlpBU
HdiZmEvWkE3MGNqQUZtY2xjeM5ZHlPRCtUanZjVINtcy9Kd0JWdmFQZFE4Q2pJZjJlcQowdEQ4YIFhYkgxL2VhNVM0THg5NFd
FZ3NjNDAYNkxac1dwVEpUSFhBdFhnSUZvbWxsaDBZUHkyNWRnMmpEbFh0CmhlaFFFQkJ2VTZWazNaaEZUalBKcWQvaUcwO
FIXV44eVNyOEN4cExKWmFCS29kT3VHaCszNHlmQTBXcE41dlgKakFRSIZRYmpqNGVWWIIQa25zUEpGcXdvL3pzSm1MazOKL
S0tLS1FTkQgUFJJVkfURSBLRVktLS0tLQo=

[student@workstation acme]\$

[student@workstation acme]\$ oc config view

apiVersion: v1

clusters:

- cluster:

server: https://api.ocp4.example.com:6443

name: api-ocp4-example-com:6443

contexts:

- context:

cluster: api-ocp4-example-com:6443

namespace: default

user: admin/api-ocp4-example-com:6443

name: default/api-ocp4-example-com:6443/admin

kind: Config

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preferences: {}

users:

- name: acme-auditor

user:

client-certificate-data:

LS0tLS1CRUdJTiBDRVJUSUZJQ0FUR50tLS0tCk1JSUVJakNDQXdxZ0F3SUJBZ0lSQU1NSkhERTZzUVR5cTVMMem53VHdOWTh3RF
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S0tLS1FTkQgUFJJVkfURSBLRVktLS0tLQo=

[student@workstation acme]\$

```
[student@workstation acme]$ oc config --kubeconfig acme-corp.config set-context acme-auditor --cluster api-ocp4-  
example-com:6443 --namespace default --user acme-auditor
```

Context "acme-auditor" created.

[student@workstation acme]\$ cat acme-corp.config

apiVersion: v1

clusters:

- cluster:

certificate-authority-data:

LS0tLS1CRUdJTiBDRVJUSUZJQ0FUR50tLS0tCk1JSU0zVENDQTBXZ0F3SUJBZ0lCSmpBTkna3Foa2lHOXcwQkFRc0ZBREJITVJRd0
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```
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```

server: https://api.ocp4.example.com:6443

name: api-ocp4-example-com:6443

contexts:

- context:

cluster: api-ocp4-example-com:6443

namespace: default

user: acme-auditor

name: acme-auditor

current-context: ""

kind: Config

preferences: {}

users:

- name: acme-auditor

user:

client-certificate-data:

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dSREFUCKJnTlZlU1VFRERBS0JnZ3JCZ0VGQlFjREFqQU1CZ05WSFJNQkFmOEVBakFBTUI4R0ExVWRJd1FZTUJhQUZOMESKT2ZT
RTFBZmFUY2krTjF2UzVJRmM0ZFpKTUeWRONTcUdTSWlZrFFFQkN3VUFBNEDQVFCRzV2SUo1elUrOUtpdgpBV1BJN3NMMXg
4T0gzemxTaTRCZ3pzN3Nzbk9FbWVoWGEweHc3T3Q0ZytPYVJjcmNaVnZmU2FnNW43T1A5c3lsCmpNZ0kxR1habGgxZ2pVSIZ
WYU8rUVRyYdWUxYmp6MFlmbzg2cFZOU1FhS1RjSmNQdmNJaFBWba2R2VkvVeTJlCkFMKY2ZTRlARgJlVzF4bWt5Y3lRV2MrQk
0wL0pjdZn1ZEY0SlhMOTdROHBsVGhpNlVWbUMvUmhzRitPSDNWcnZVZworTEdhemQ2SUIvNkp2QWhwT2hvVjNXaWMxVm
wzbXpyWGVocEt2aWh5aVhpRzliK0o1UVN6Wk1RTUU1N3JweW1yCitoVUdibXhycUFQbTVGZWcyZ2ZCa3JCRVZQV2g1aUNDZ
XhNeU13N0FIS283Q21EdzY5eHgwTHVvc1BnTy9hWlQKQ1g2ZktjRW4KLS0tLS1FTkQgQ0VSVElGSUNBVEUtLS0tLQo=
```

client-key-data:

```
LS0tLS1CRUdJTiBQUklWQVRFIEtFW50tLS0tCk1JSUpRd0lCQURBTk1na3Foa2lHOXcwQkFRRUZBQVNDQ1Mwd2dna3BBZ0VBQ
W9JQ0FRQ3pwOGHeelhrUHHdZTKKcFF5TGJSdGkwdHptVkkzd1ZkaVByNmC0dlVYz3VzcE8vZWpPZXl3UTBweDVRMmtSZFpES
3IPOUhCdUt2eDBpQQp5U2o4WGpHeUJCejl2bFhoSlhGVUM1Y2NWYytUSnI1WWlwbENEeDNXOElyakNvTk9FTnhaQzIEUnIzR
GpKcVJ1CkVDDTQrSFIFZnZFKzlyZHBlamFJV2VvQytZTDNQYWNZOuhjdGl5czdvcGJuU1ZPTWVyd1h4MGpXVWkzK2p2OEMKZ3
IJVTBvRFNpdTE1VFljWjRKTDFtbi9wcDhVK3MrcHJXYWVsYUUVFakiOemNaTHdDd09PRW9kTkJnRDFXcVpHdQpBVDRJcFZkY09u
TXk1QU50Z1FqekdmZklrK05vSXJtamRHcXAMTDEMGP0ZEFyR3dhdHdkMlVHRUxhTXYwNkc3Cmw1YlIRUzZUU09BdVBaaVNI
MkJ5ZytrZGpVWTVVMmJ3b1E5czFYcTNiaTZSVzZPTTRJbEQyN2VYWGJ6RkI3TkWKejlZdWlaMUdMTUh6L2NzS3hxVDFQQWQ0
NkJEUxIKMEx1KzRzZktpZU95Q2hkbExleHIUdDVhQkpCRXk0ZWVONApPakM5Uk1Bd0Z2My81MkhMMnBPbTBSY2xhWkp4UD
```

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JQb0V6ODJ0SWpnMmw2cE9Cc0VVMXRLYXV1QWRyUHBZNDFvCIZYNWd2K1N6NUxbzNySytXdFRFM2FtUXh4Uytpc2RwRFE
2YzF2OEs0RCtqUzIjVWtHbzF2KytkTDR0ZHg0cG0KSEN4RVFWaIFpMnVHdmgwN1czOTZmQWk0bDFqRS9vcWZxTEMwbWhEV
VBudXNMQkFoY1pnY3d3TE5ESzZRVjJXVwpCdHFTOGZKUjdmWEJFSkZOMjVmaXVBdnFKMG5TT3dJREFRQUJBB0IDQUFiSmRn
b3RzZfJKSmthR3A0aUFPTjBpCllkV252Y0dsdjBFcVhKSUs2MUJvYmhVa01zNStIbVkb1ZGVUHSemJrcVUraUYrU3RkbjF6dG8zL3
o3aXkydDgKdHFBVfDudZJDYjBtNmtZQ21JMW9GTXAxZXIYWTVBUGNqQ1ZUU1M1Q3BTTU9ER0pUTGtzbWIMd0p0Z1VqdEh
NQQpzcTI4RFhhUmR6YnVkl3MxeVluNmxSeWZFUDR5VFPInJLQnAzT2luM2F3NDIik2hlaVd0bEVVQzNEVHFLNU5kCjIwRUlH
ZXdETFE5c05EOHpweTF5WnVRNFVRVGNLk2h5dEJKdFp5R08xT21MK0NyM2VSbSszVEhlejgraGVNRSsKQ2dwb1RSZ3V4dWp
4Z3dnR1RPY3JES3FpcEJiczBTS0NYSTROSXl1cTFjeExpXmVZlhtUFFRTVJFU2RKU3hwQQpETm5ZOGVhZWtpRS93dFhmclVvbW
xLeEU1SWttcEN1eXI0ckxmSHZ3ampNbHhoT0ZQcXpiRjF2aIFiTHFEQVQvCnVDY0drRDRFYjBNcU9reXBpQkFRb25ydUNkeks2a
U5TSIIEQWtQM1NKZ21EN0JwVy9wdjV0NIBIVHdCQ2R3TW8Kb21ubjZNUtRHZCtna2Z6MVF5VnIIRTByZ2VYbTdobIRqUVhMN
TdnNlovc3VxcDhkUEdBTUpwZjY4eVN4Y0F5aAp3QksrRnB1YU13a2lpNHplYXIYm2xEOdIwejdRsk13SVhHRG1PQW5sVExucktx
cG9NRjJ2RENHdmpUTDhkVzFpCnlWUUFM4NHpZSEpkSkxpOUYzd0pzVncyTEY2Y1I5VWZocGlqU1oxMEXLRkpMRzVpSU5wRU0
xd1sdEV6dmsvYlckS1RKQ2QzS25EZWpuZzdZOE NmSnBBb0lCQVFERDNLRG5JRXdON09BMDZyQW1kMDNNT2RYdDVjN3FQb
1FzaFZBTApbjY3OFcyVndBdmptQitIVnR0OUxnRWVak256eW9mR3M5THN6dkdpU0JET0V3eDFNQINvL09maU4yK3luaTYyC
mpEeWIEZ05odEFwZjQ2bXF5UXFuVElIUU9IMVE2YWR3NFZ5Nm01bFpCRGNDMWIHYzygbFRVOWFhbWxMYXpMRmQKbmZ
QOXDZQXIsNW1qWWtoSGtiTy8yS3VVbVZBYWV3K3k3ZkV4ekRQL1BiWEhwMnhoa0ZBNVUzRU1qNWFhaWFqTApzSi9GTmt1
NmJXUzVkrFEvQ1BXUzIzeU9OVEVjb1dwVWdEM3F2RGRIVVBhdUtNcjJ6dGIIYjZnSzdrRjTbXpUCjRGNU9VaHpsVitKU0huWEJ
qb2RVSzRPK3FTVjNRdkVoNUK5UIE0VFNhY2ILRGcyakFvSUJBUURxMFVhckJtUEQKbFVPUVdkR1BXOFNjaStLdlcySEtma0V5SUZ
zdFE2ZnRyRzFPN0lrTDZGcXFENEw5NVJSTIB5VGNvZDVsekxGMwpCcXVrb0NQKk14T2M1MUNlcG5uVFRVS3pnZXlxejl4MFFh
d01pUzJXcFhJKzNjQUtUbXfmbVrtZi9JclJ6WW9rCm9TeGtMTUxQUzIOQUlZr25xa3QwMWtPRWlsVVh4TEJNTkViaktmN3VxM
HhpSjlqQXJ1VTkxR01SMXNBuE9yNUSKYXRBk5pR056bkV4R21oaXM5Qzk0TkJEctFYl3Y5WFJBOWhpjdhVbXduDxdWV0p3R
1JRTFF3RGRYdGI4UTVaNAo1MDY0Y09ZdVhmRWFDYVpXVkJuSmDKU0RPSDBJY2tWVksrNFIUbjlyZUNPKzd3ZT21RXIVbk1PST
RFeUwzMC9ECngxNWFWeEpyeGNLSkFvSUJBUUREYVJvM3ZxZGI0VIMwTnZwc1JsZWVUSTI5L3F4RVZQODJGQVBhcjllRjE4aEg
Kd2dXYW15bkpzOFRemNOYi9MMFQRsnNpaWtNRTITdHJoNDhNcTBDZ29zbmFHQW1hVDBqbjRwNGtqeUhuWINmRwprUVJ
ReXIsNk5YNmxqOVZzVkg4TFhUTXJrZzBJT1NYMEJCcUdmL0dsVDGxano1K3JQKkV5TDBzK1pkc0s3OFhoCnRCTWJTWEZueGx5Y
UthaDJaOGZkdZyZVURVVzVNMDBSZVFjBUlrb2x4UC9sVWZUV1hXUWE1d0U3eUFwL2pqNWIKejh6UW9ybUZzb0NnZHM5OG
FtR0hIUnhIRm5Pd0FQMfU0cDViQUdRUGZpdldaSFk2QVovYVQwTG8yVS85d0RzSQpMbmlsQ2JMZTNENzBvVDJIT2RBaGZ1NX
ByRW01QUM1Q2FEd21VUU1WQW9JQkFRRGw4NmXqEc4Z1VtVUcwSjZpCjZEaERJYzRNOGMxelVtNmdHd1dPN3IsZFhEVHp
TOUtmJ1I4a1dPcHJudeZ0MEV0VjhrSnM2S2FGckRIY2tPSIMKeWfVVM0WVdRVIIITWhQMjNHNGEzRkhFVzZIVDR5YU0xMXB
vN0JtclBoMjJTNFFIQT4aWxFa0xFL2IzNXNtagozVnB6NHBAUG5qb2tvY3NIMmFUNUVMbWc5clRRc3dpZIM0c1N1SGpRUUxY
bjlieEs5T1hraXBoZElmcXlqODhPCKlyUEFaVjhdQVp1U3hyek05OVhSRFZqalJnTzZ1SmZtR3VkTUNHSE02eEI3OFIzQXE2YVhCYz
B1ZXo0ZzZ4YIAKSTFuL3FLR0creld1V2NIV05WUmtXQnREbGxtbC9nRIVDaVBmVUE5ZXNDcjJnZ29PV2JFOGJLU2RVZms1T1Btb
QpKdFVaQW9JQkFHZTRkeTc2eGhpY0dqSWNkRDZwYkRic2JwK3pkRnJNUXFQK3dIZkwzOGdIT0E3eFdPdInlcXIVCkE4YUwwRis
1WkNGTmNLdWlqbUJBV0hzenVYNzBBEwJzcDlhZWWhnaThMWGIQYIJMR0VktX5d5dTzenkyZnFhZDYKWfhdTNCdnkyTlpBU
HdiZmEvWkE3MGNqQUZtY2xjeM5ZHLpRctUanZjVINtcy9Kd0JWdmFQZFE4Q2pJZjJlcQowdEQ4YIFhYkgxL2VhNVM0THg5NFD
FZ3NjNDAYNkxac1dwVEpUSFhBdFhnSUZvbWxsaDBZUHkyNWRnMmpEbFh0CmhlaFFFQkJ2VTZWazNaaEZUalBKcWQvaUcwO
FIXV44eVNyOEN4cExKWmFCS29kT3VHaCszNHlmQTBXcE41dlgKakFRSIZRYmpqNGVWWIIQa25zUEpGcXdxL3pzSm1Maz0KL
S0tLS1FTkQgUFJJVkfURSBLRVktLS0tLQo=

```
[student@workstation acme]$ oc config use-context acme-auditor --kubeconfig acme-corp.config  
switched to context "acme-auditor".
```

```
[student@workstation acme]$ oc whoami --kubeconfig acme-corp.config  
acme-auditor
```

```
[student@workstation acme]$ oc auth can-i get pods -A --as acme-auditor --as-group acme-corp
```

Yes

```
[student@workstation acme]$ oc auth can-i create pods -A --as acme-auditor --as-group acme-corp
```

No

Q3: Restore an application from a backup

The application <https://lynx.apps.domain21.example.com> was mistakenly deleted.

Restore the application per the following requirements:

- The latest completed scheduled backup is used to create a restore called **restore-lynx-web**

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- The application <https://lynx.apps.domain21.example.com> is accessible using the credentials **admin/Admin1andia!1958**

Solution

To restore the application <https://lynx.apps.domain21.example.com> using the latest completed scheduled backup and ensure it is accessible with the credential's admin/Admin1andia!1958

check oadp and volSync operators installed

```
[student@workstation backup-restore]$ oc get og,ip,sub,csv -n openshift-adp
```

do restore from current backup

```
[student@workstation backup-restore]$ oc get deployment -n openshift-adp  
openshift-adp-controller-manager
```

```
velero
```

```
[student@workstation backup-restore]$ alias velero='oc exec -it -n openshift-adp deployment/velero -c velero -- ./velero'
```

```
[student@workstation backup-restore]$ velero get backup
```

```
restore-lynx-web
```

```
[student@workstation backup-restore]$ velero describe backup restore-lynx-web | grep -i namespace  
lynx-web
```

```
[student@workstation backup-restore]$ velero get backup restore-lynx-web | grep -i namespace -A 3  
lynx-web
```

```
[student@workstation backup-restore]$ oc project lynx-web
```

```
[student@workstation backup-restore]$ vim restore.yml
```

```
apiVersion: velero.io/v1
```

```
kind: Restore
```

```
metadata:
```

```
  name: lynx-web
```

```
  namespace: openshift-adp
```

```
spec:
```

```
  backupName: restore-lynx-web
```

```
  restorePVs: true
```

```
  namespaceMapping:
```

```
    lynx-web: lynx-web
```

```
:wq!
```

```
[student@workstation backup-restore]$ oc apply -f restore.yml
```

```
[student@workstation backup-restore]$ velero get restore
```

```
[student@workstation backup-restore]$ oc project
```

```
[student@workstation backup-restore]$ oc get route
```

Verify application through route address and credentials

in Lab to practice

Backup and Restore with OADP -> page 186

Q4: Deploy a Machine Configuration using GITOPS

Deploy the OpenShift GitOps operator according to the following requirements:

- The operator is installed in the openshift-gitops-operator project.
- The ArgoCD instance deployed by the operator has **TLS** enabled with **reencrypt** termination.
- The **cluster-admin** user is a member of the **gitops-admins** group.
- The **gitops-admins** group is the only group configured as **role:admin** with RBAC key policy.
- An ArgoCD instance Git repository exists at <http://git.domain21.example.com/ocp-gitops> with skip server verification enabled.

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- An application named **machineconfig-motd-deploy** exists in the Git repository with the Sync Policy configured as **Manual**.

The Git repository <http://git.domain21.example.com/ocp-gitops.git> contains two MachineConfig resources as follows:

A file called **71-master-sshd-motd.yaml** which uses the latest ignition version with the following configuration:

- A path exists called **/etc/motd**
- The filesystem is **root**
- The overwrite policy is **true**
- The object is applied only on the nodes marked with **machineconfiguration.openshift.io/role: master**

A file called **71-worker-sshd-motd.yaml** which uses the latest ignition version with the following configuration:

- A path exists called **/etc/motd**
- The filesystem is **root**
- The overwrite policy is **true**
- The object is applied only on the nodes marked with **machineconfiguration.openshift.io/role: worker**

In both cases, the machine configuration:

- Create a file located at <http://rhgls.domain21.example.com/ssh.motd>
- Sets the permission on **/etc/motd** on all nodes to **r - - r - -**

Solution

```
[student@workstation ~]$ oc whoami
admin
```

```
[student@workstation ~]$ oc whoami --show-console
```

<https://console-openshift-console.apps.ocp4.example.com>

Login in to OCP GUI and check if the git operator installed or not. If not, install it first.

After install, verify:

```
[student@workstation ~]$ oc get og -n openshift-gitops-operator
```

```
NAME                                AGE
openshift-gitops-operator-xt79f    2m
```

```
[student@workstation ~]$ oc get ip -n openshift-gitops-operator
```

```
NAME      CSV                                APPROVAL APPROVED
install-dprmh  openshift-gitops-operator.v1.10.1 Automatic true
```

```
[student@workstation ~]$ oc get sub -n openshift-gitops-operator
```

```
NAME                                PACKAGE                                SOURCE                                CHANNEL
openshift-gitops-operator            openshift-gitops-operator              do380-catalog-cs                    latest
```

```
[student@workstation ~]$ oc get csv -n openshift-gitops-operator
```

```
NAME                                DISPLAY                                VERSION REPLACES                                PHASE
openshift-gitops-operator.v1.10.1  Red Hat OpenShift GitOps              1.10.1                                openshift-gitops-operator.v1.10.0
Succeeded
```

Next, config ArgoCD default instance:

```
[student@workstation ~]$ oc adm groups new gitops-admins
```

```
group.user.openshift.io/gitops-admins created
```

```
[student@workstation ~]$ oc get groups.user.openshift.io
```

```
NAME      USERS
```

```
Default SMB Group
```

```
gitops-admins
```

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```
station ~]$ oc adm groups add-users gitops-admins cluster-admin
group.user.openshift.io/gitops-admins added: "cluster-admin"
```

```
[student@workstation ~]$ oc get groups.user.openshift.io gitops-admins -oyaml
apiVersion: user.openshift.io/v1
kind: Group
metadata:
  creationTimestamp: "2025-01-21T10:13:10Z"
  name: gitops-admins
  resourceVersion: "296842"
  uid: 1fe883ca-ce39-43e9-9830-ba6e832ccfe0
```

```
users:
- cluster-admin
[student@workstation ~]$ oc edit argocds.argoproj.io openshift-gitops
route:
  enabled: false
  tls:
    termination: reencrypt
```

```
rbac:
  defaultPolicy: ""
  policy: |
    g, system:cluster-admins, role:admin
    g, cluster-admins, role:admin
    g, ocpadmins, role:admin
    g, gitops-admins, role:admin
```

```
:wq!
```

```
[student@workstation ~]$ oc create configmap -n openshift-gitops cluster-root-ca-bundle
```

```
[student@workstation ~]$ oc label configmaps -n openshift-gitops cluster-root-ca-bundle config.openshift.io/inject-trusted-cabundle=true
```

```
[student@workstation ~]$ oc get argocd
```

```
repo:
resources:
  limits:
    cpu: "1"
    memory: 1Gi
  requests:
    cpu: 250m
    memory: 256Mi
volumeMounts:
- name: cluster-root-ca-bundle
  mountPath: /etc/pki/ca-trust/extracted/pem/tls-ca-bundle.pem
  subPath: ca-bundle-crt
volumes:
- configMap:
    name: cluster-root-ca-bundle
  name: cluster-root-ca-bundle
```

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resourceExclusions: |

:wq!

Now login to git

<https://git.ocp4.example.com>

Create new project ->ocp-gitops and make sure its accessible PUBLIC

\$ pwd

/home/student

\$ mkdir git

\$ cd git

\$ echo "ex380v" >motd

[student@workstation git]\$ oc get mc 99-master-chrony-conf-override -oyaml >71-master-sshd-motd.yaml

[student@workstation git]\$ oc get mc 99-worker-chrony-conf-override -oyaml >71-worker-sshd-motd.yaml

student@workstation git]\$ cat motd | base64 -w0;echo

ZXgzODB2NC4xNAo=

[student@workstation git]\$ cat 71-master-sshd-motd.yaml

apiVersion: machineconfiguration.openshift.io/v1

kind: MachineConfig

metadata:

labels:

machineconfiguration.openshift.io/role: master

name: 71-master-sshd-motd

spec:

config:

ignition:

version: 3.2.0

storage:

files:

- contents:

source: data:;base64,ZXgzODB2NC4xNAo=

mode: 444

overwrite: true

path: /etc/motd

[student@workstation git]\$ cat 71-worker-sshd-motd.yaml

apiVersion: machineconfiguration.openshift.io/v1

kind: MachineConfig

metadata:

labels:

machineconfiguration.openshift.io/role: worker

name: 71-worker-sshd-motd

spec:

config:

ignition:

version: 3.2.0

storage:

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files:

- contents:

source: data:;base64,ZXgzODB2NC4xNAo=

mode: 444

overwrite: true

path: /etc/motd

Clone git repository

```
[student@workstation git]$ git clone https://git.ocp4.example.com/developer/ocp-gitops.git
```

```
[student@workstation git]$ ls
```

```
71-master-sshd-motd.yaml 71-worker-sshd-motd.yaml motd ocp-gitops
```

```
[student@workstation git]$ cp 71* ocp-gitops/
```

```
[student@workstation git]$ cd ocp-gitops/
```

```
[student@workstation ocp-gitops]$ ls
```

```
71-master-sshd-motd.yaml 71-worker-sshd-motd.yaml README.md
```

```
[student@workstation ocp-gitops]$ git add *
```

```
[student@workstation ocp-gitops]$ git commit -m "first push"
```

```
[student@workstation ocp-gitops]$ git push
```

Check git repo for changes, refresh the page

You can watch 71-.... Files in your git address

Now, its ArgoCD time,

create an ArgoCD application

follow exam orientation video recording

Q5: Configure LOG Forwarding

Configure the OpenShift-logging project in your OpenShift cluster to forward logs to an external aggregator according to the following specifications:

- The log collector type is **vector**
- **Application** and **infrastructure** logs are forwarded to the syslog service on host **log.domain21.example.com**
- **Audit** logs are forwarded to the syslog service on host **vault.domain21.example.com**
- The syslog service on both hosts listens on port **tcp/514** and **udp/514**
- Application logs are tagged with **procID: app**
- Infrastructure logs are tagged with **procID: infra**
- Audit logs are tagged with **procID: audit**
- All logs are tagged with **appName: openshift**

The Event Router is running using the registry image:

- **registry.domain21.example.com/openshift-logging/eventrouter-rhel9:v0.4**

The syslog servers have already been fully configured for you to receive logging messages and filter them using the **procID** attribute to the following files:

- **/var/log/ex380-app.log**
- **/var/log/ex380-infra.log**
- **/var/log/ex380-audit.log**

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You may check your work by logging into the syslog servers using the **opsadm** user to examine the logs. You do not have root access to these systems.

Solution

Install the OpenShift Logging Operator

If the OpenShift Logging Operator is not already installed, install it from the OperatorHub in the OpenShift web console.

Create a ClusterLogging Custom Resource (CR)

```
[student@workstation ~]$ vim clusterlogging.yaml
apiVersion: logging.openshift.io/v1
kind: ClusterLogging
metadata:
  name: instance
  namespace: openshift-logging
spec:
  managementState: Managed
  logStore:
    type: ""
  visualization:
    type: ""
  collection:
    logs:
      type: vector
      vector:
        resources: {}
:wq!
[student@workstation ~]$ oc apply -f clusterlogging.yaml
[student@workstation ~]$ oc get clusterlogging
```

Create a ClusterLogForwarder Custom Resource (CR)

```
[student@workstation ~]$ vim clf.yaml
apiVersion: logging.openshift.io/v1
kind: ClusterLogForwarder
metadata:
  name: instance
  namespace: openshift-logging
spec:
  outputs:
    - name: app-syslog
      type: syslog
      url: 'tcp://log.domain21.example.com:514'
      syslog:
        appName: openshift
        proclD: app
    - name: infra-syslog
      type: syslog
      url: 'udp://log.domain21.example.com:514'
      syslog:
        appName: openshift
        proclD: infra
    - name: audit-syslog
```


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```
type: syslog
url: 'tcp://vault.domain21.example.com:514'
syslog:
  appName: openshift
  proclD: audit
```

```
pipelines:
- name: app-logs
  inputRefs:
    - application
  outputRefs:
    - app-syslog
- name: infra-logs
  inputRefs:
    - infrastructure
  outputRefs:
    - infra-syslog
- name: audit-logs
  inputRefs:
    - audit
  outputRefs:
    - audit-syslog
```

:wq!

```
[student@workstation ~]$ oc apply -f clf.yaml
```

```
[student@workstation ~]$ oc get clf
```

Deploy the Event Router

```
[student@workstation ~]$ oc create deployment eventrouter --image registry.domain21.example.com/openshift-logging/eventrouter-rhel9:v0.4 -oyaml --dry-run=client >eventrouter.yaml
```

```
[student@workstation ~]$ vim eventrouter.yaml
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: eventrouter
```

```
  namespace: openshift-logging
```

```
spec:
```

```
  replicas: 1
```

```
  selector:
```

```
    matchLabels:
```

```
      app: eventrouter
```

```
  template:
```

```
    metadata:
```

```
      labels:
```

```
        app: eventrouter
```

```
    spec:
```

```
      containers:
```

```
        - name: eventrouter
```

```
          image: registry.domain21.example.com/openshift-logging/eventrouter-rhel9:v0.4
```

```
          args:
```

```
            - -logtostderr
```

```
            - -v=2
```

:wq!

EX380v4.14-Certified Specialist in OpenShift Automation and Integration

```
[student@workstation ~]$ oc apply eventrouter.yaml
[student@workstation ~]$ oc get eventrouter
[student@workstation ~]$ oc get pods -n openshift-logging
[student@workstation ~]$ oc logs -n openshift-logging -l app=eventrouter
[student@workstation ~]$ ssh opsadm@log.domain21.example.com "tail -f /var/log/ex380-app.log"
[student@workstation ~]$ oc get clusterlogforwarder -n openshift-logging
[student@workstation ~]$ oc get clusterlogging -n openshift-logging
[student@workstation ~]$ oc logs -n openshift-logging -l component=collector
```

Q6: Troubleshooting POD Scheduling

A work colleague accidentally misconfigured your OpenShift cluster prior to leaving on vacation.

As a result, application pods are not able to run. Diagnose and correct the problem so that application pods are able to run on any worker node.

You may use the image from -> registry.domain21.example.com/openshift/hello-openshift to verify your work.

The image must deploy into a Running container without intervention.

Solution

check all workerNodes for **taint** and remove it.

```
[student@workstation ~]$ oc get nodes worker01 -oyaml | grep -i taint -A 4
[student@workstation ~]$ oc get nodes worker02 -oyaml | grep -i taint -A 4
[student@workstation ~]$ oc get nodes worker03 -oyaml | grep -i taint -A 4
taints:
```

- effect: **NoSchedule**

key: **key1**

value: **value1**

```
[student@workstation ~]$ oc adm taint node worker01 key1-
[student@workstation ~]$ oc adm taint node worker02 key1-
[student@workstation ~]$ oc adm taint node worker03 key1-
node/worker01 untainted
[student@workstation ~]$ oc get node worker01 -oyaml | grep -i taint -A 4
[student@workstation ~]$ oc get node worker02 -oyaml | grep -i taint -A 4
[student@workstation ~]$ oc get node worker03 -oyaml | grep -i taint -A 4
```